UNIT 4

Quantum Communication and Computing - Theoretical Perspective

Unit 4 at a glance

Quantum vs classical information · basics of quantum communication · QKD (BB84, E91) · entanglement in communication · quantum internet · intro to quantum computing · quantum parallelism · classical vs quantum gates · decoherence & error correction · real-world importance & future

1. Quantum vs Classical Information (Recap)

Classical	Quantum
Bit: 0 or 1	Qubit: $\alpha \mid 0\rangle + \beta \mid 1\rangle$
Can be copied, amplified, read many times	Cannot be cloned; reading disturbs it
Security rests on hard maths (RSA)	Security rests on laws of physics
Channel = wire, fibre, radio	Channel carries photons, preserving superposition
Noise → bit errors	Noise → bit & phase errors, decoherence

2. Basics of Quantum Communication

Quantum communication transfers quantum states (usually of photons) between distant parties, using the principles of superposition, measurement disturbance, no-cloning and entanglement. It enables tasks impossible classically - secure keys, teleportation, networking of quantum computers.

» Components of a quantum communication link

- *Source: produces single photons or entangled photon pairs (e.g. spontaneous parametric down-conversion, quantum dots).*
- *Encoder: writes information into polarization, phase, or time-bin.*
- *Quantum channel: optical fibre (loss ~0.2 dB/km at 1550 nm) or free space / satellite link.*
- *Classical channel: ordinary authenticated channel for basis comparison, error correction.*
- *Detector: single-photon detectors (APD, superconducting nanowire).*
- *Quantum memory & repeaters: store and relay quantum states over long distances.*

★ *Why photons?*

Move at light speed, barely interact with the environment (low decoherence), travel through fibre/air/space, easy to encode and measure. Drawback: they get lost (absorbed/scattered) - and a lost photon cannot be re-amplified because of no-cloning.

» Channel loss problem

Fibre loss grows exponentially: after 100 km (0.2 dB/km) only 1 % of photons survive; after 500 km only 10^{-10}. Classical signals are amplified every ~80 km; quantum signals cannot be amplified (no-cloning). Solutions: quantum repeaters (entanglement swapping + quantum memory + purification) or satellites (loss mostly in the first few km of atmosphere).

3. Quantum Key Distribution (QKD)

Secure communication needs a shared secret key. Classical key-exchange (RSA, Diffie-Hellman) relies on the assumed difficulty of factoring - broken by Shor's algorithm on a large quantum computer. QKD lets two parties (Alice & Bob) establish a random secret key whose security is guaranteed by quantum physics, and any eavesdropper (Eve) is detected.

» 3.1 BB84 protocol (Bennett & Brassard, 1984)

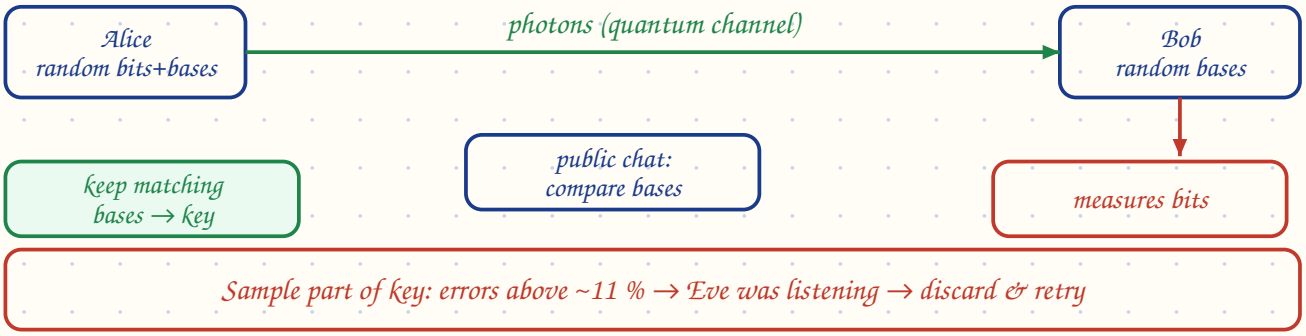


Fig 1: BB84 quantum key distribution

1. *Alice generates random bits and, for each bit, a random basis: rectilinear (+: H/V) or diagonal (×: D/A).*
2. *She sends single photons: bit 0/1 in + basis → H/V; bit 0/1 in × basis → D/A.*
3. *Bob measures each photon in a randomly chosen basis. If he picks the wrong basis, result is random.*
4. *Over a public channel they announce the bases (not the bits) and keep only positions where bases matched (~50 %). This is the sifted key.*
5. *They publicly compare a random sample. If the error rate (QBER) is low (< ~11 %), no significant eavesdropping; else abort.*
6. *Error correction + privacy amplification → final secret key; use it with one-time pad (perfect secrecy) or AES.*

Alice bit	1	0	1	1	0
Alice basis	+	×	×	+	×
Photon sent	V	D	A	V	D
Bob basis	+	+	×	×	×
Bob result	1	random	1	random	0
Keep?	✓	✗	✓	✗	✓

If Eve intercepts and measures in a random basis, she guesses wrong 50 % of the time; she then resends a wrong-state photon, which makes Bob's result wrong in 50 % of those cases → 25 % error in the sifted key. Eve is detected by the error rate. She cannot copy photons (no-cloning).

Other QKD protocols

B92: two non-orthogonal states. E91 (Ekert, 1991): uses entangled pairs; security from Bell-inequality violation. Decoy-state BB84: fixes weak-laser-pulse photon-number-splitting attacks (practical). CV-QKD: continuous variables, uses standard telecom components. MDI-QKD / Twin-field: remove detector vulnerabilities and extend distance.

» Practical issues

- Real sources are weak lasers, not perfect single-photon sources → decoy states.
- Detector side-channel attacks (blinding) → device-independent / MDI protocols.
- Distance limited to ~100-400 km in fibre without repeaters; key rate falls with distance.
- Needs authenticated classical channel (pre-shared short key).
- Standards bodies (ETSI, ITU, ISO) and NSA/NCSC caution: QKD complements, but does not replace, post-quantum cryptography.

Exam tip / Remember

Be able to (i) draw BB84 flow, (ii) work a table of bits/bases/sifted key, (iii) explain why eavesdropping causes 25 % errors, (iv) mention no-cloning and measurement disturbance.

4. Role of Entanglement in Communication

- E91 QKD: a source sends one photon of each entangled pair to Alice and Bob. Matching-basis results are perfectly correlated → key. Bell test on remaining data certifies no eavesdropper (device-independent security).
- Quantum teleportation: transmit an unknown qubit using shared entanglement + 2 classical bits (see Unit 2). Basis for quantum networks and gate teleportation in computers.

- *Entanglement swapping*: if A-B and C-D are entangled pairs, a Bell measurement on B and C entangles A and D even though they never met. This is the core of a quantum repeater.
- *Superdense coding*: 2 classical bits per qubit using entanglement.
- *Entanglement distribution*: satellites (Micius: 1200 km), fibre links (Delft 1.3 km loophole-free; Chinese/Euro networks).
- *Distributed quantum computing / sensing*: entangled links let separate processors act as one.

> **Quantum repeater idea**
> Divide long distance into short segments. Create entanglement in each segment → store in quantum memories → purify (distil better entanglement from noisy pairs) → connect segments by entanglement swapping. Loss then grows only polynomially, not exponentially.

5. The Idea of the Quantum Internet

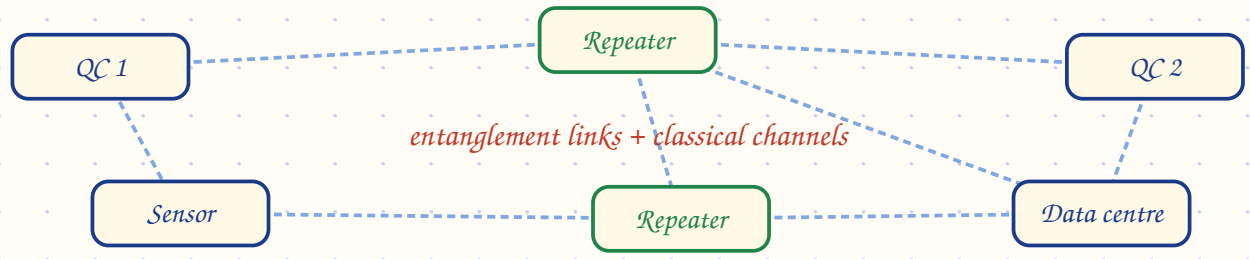


Fig 2: Sketch of a quantum internet

A *quantum internet* is a network that can distribute entanglement and transmit qubits between quantum devices anywhere in the world, running alongside the classical internet (which is still needed for control signals). Kimble (2008) laid out the vision; Wehner, Elkouss and Hanson (2018) described stages of its development.

Stage	Capability	Status
1. Trusted-repeater networks	QKD between nodes (nodes trusted)	Operational (Beijing-Shanghai, SECOQC, Tokyo QKD, India pilot links)
2. Prepare-and-measure networks	End-to-end QKD & secure comms without trusting nodes	Lab demonstrations
3. Entanglement distribution	Entangled pairs on demand between any two nodes	Early experiments (Delft, Hefei)
4. Quantum-memory networks	Storage, teleportation of qubits	Research
5. Few-qubit fault-tolerant networks	Distributed quantum computing, blind computing	Future

Stage	Capability	Status
6. Quantum computing networks	Full-scale network of QCs	Long-term

» Applications of the quantum internet

- Provably secure communication & key distribution at global scale.
- Secure access to cloud quantum computers (blind quantum computing - server cannot see your data or algorithm).
- Distributed quantum computing: link small processors to make a big one.
- Clock synchronization and networks of telescopes / sensors with improved precision.
- Quantum voting, secret sharing, secure multi-party computation, quantum money.

> Important projects
> India: ISRO free-space QKD demos (300 m → 1 km), DRDO-IIT Delhi 100 km QKD, NQM satellite & inter-city targets. China: Micius satellite, 2000 km backbone. EU: EuroQCI. USA: DOE quantum internet blueprint (2020), Chicago network. Netherlands: QuTech Delft - Hague link. Singapore, Japan, South Korea, UK networks.

6. Introduction to Quantum Computing

A quantum computer uses qubits and quantum gates to process information. The main idea (Feynman 1981, Deutsch 1985): since nature obeys quantum mechanics, a computer built from quantum components can simulate nature - and some mathematical problems - far more efficiently than classical computers.

- Input state is prepared in $|00\ldots0\rangle$.
- A sequence of quantum gates (unitary operations) creates superpositions, entanglement and interference.
- Finally, qubits are measured to obtain a classical bit string; algorithm is repeated many times to estimate the answer's probability.

> Famous algorithms
> Shor's algorithm (1994): factoring in polynomial time → breaks RSA/ECC. (Exponential speed-up.)
> Grover's algorithm (1996): search among N items in $\sim\sqrt{N}$ steps (quadratic speed-up).
> Quantum simulation / VQE / QPE: chemistry & materials.
> HHL: linear systems; QAOA: optimization; Quantum machine learning (under study).
> Deutsch-Jozsa, Simon: early proofs of quantum advantage.

7. Quantum Parallelism (Many States at Once)

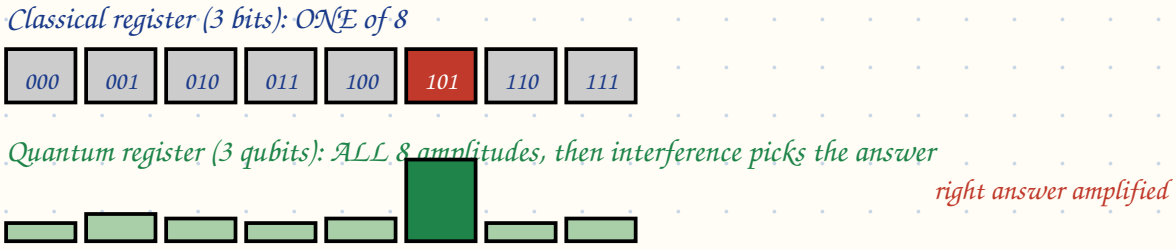


Fig 3: Superposition of all inputs - then interference

Apply Hadamard gates to n qubits ($|0\rangle^{\otimes n}$) → equal superposition of all 2^n inputs: $(1/\sqrt{2^n})$ $\Sigma\,|x\rangle$. Applying a function $f(x)$ implemented as a unitary U_f acts on every term simultaneously: $\Sigma\,|x\rangle\,|f(x)\rangle$. This is quantum parallelism.

★ *But there is a catch!*

Measurement returns only ONE random x and $f(x)$. The advantage comes from interference: algorithms are designed so wrong answers cancel and right ones add. Without clever interference there is no speed-up. Quantum parallelism ≠ 'brute force on all cases and read all answers'.

Deutsch's problem (smallest example)

Given $f:\{0,1\}\to\{0,1\}$, is f constant or balanced? Classically: 2 evaluations. Quantum: 1 evaluation (using superposition and interference) - the first demonstration that a quantum algorithm can beat classical.

Problem	Best classical	Quantum	Speed-up
Factoring N	sub-exponential (GNFS)	Shor: polynomial in log N	Exponential (superpolynomial)
Unstructured search (N items)	$O(N)$	Grover: $O(\sqrt{N})$	Quadratic
Simulating quantum systems	Exponential in particle number	Polynomial	Exponential
Sorting / basic arithmetic	Fast already	No advantage	None

8. Classical Gates vs Quantum Gates

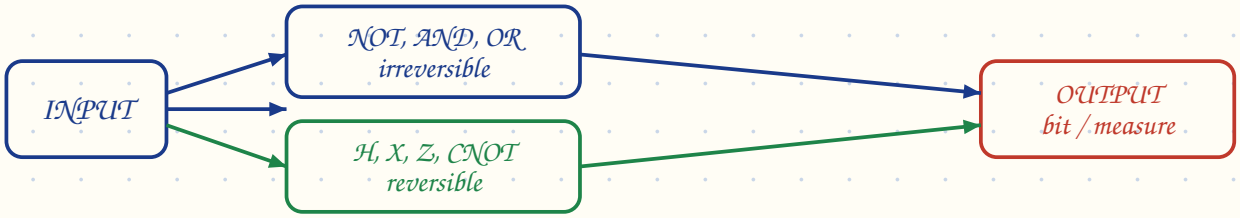


Fig 4: Irreversible classical logic vs reversible quantum gates

Classical gate	Quantum gate
NOT, AND, OR, NAND, XOR	X, Y, Z, H, S, T, CNOT, CZ, Toffoli
Irreversible (AND loses information, generates heat)	Reversible (unitary); no information lost
Acts on definite bit values	Acts on superpositions linearly
Truth table	Unitary matrix
NAND is universal	H + T + CNOT is universal (approximately)
Output always definite	Output is a state; measurement needed
Can copy bit (fan-out)	Cannot copy qubit (no-cloning)

» Important quantum gates

Gate	Effect	Matrix
X (NOT)	$\|0\rangle \leftrightarrow \|1\rangle$ (bit flip)	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Z	$\|1\rangle \rightarrow -\|1\rangle$ (phase flip)	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Y	bit and phase flip	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
H (Hadamard)	$\|0\rangle \rightarrow (\|0\rangle + \|1\rangle)/\sqrt{2}$, $\|1\rangle \rightarrow (\|0\rangle - \|1\rangle)/\sqrt{2}$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$
S, T	phase shifts of 90°, 45°	$\mathrm{diag}(1,i)$, $\mathrm{diag}(1,e^{i\pi/4})$
CNOT	Flip target if control = 1 (entangles)	4×4 matrix
Toffoli (CCNOT)	Flip target if both controls = 1	8×8; makes classical logic reversible

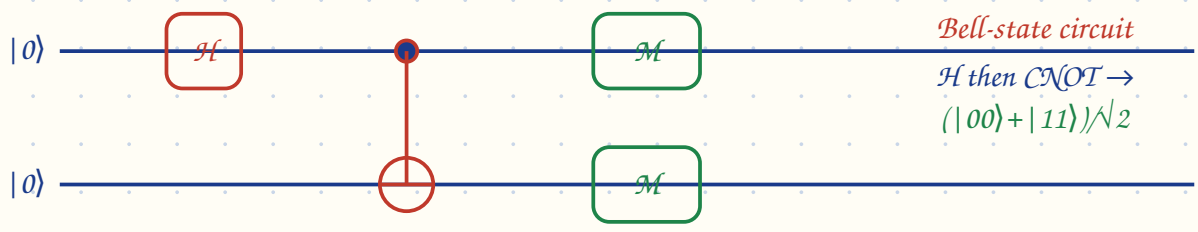


Fig 5: Circuit that creates a Bell state

Creating a Bell state

Start $|00\rangle \rightarrow H$ on qubit 1: $(|0\rangle + |1\rangle)|0\rangle/\sqrt{2} \rightarrow$ CNOT (control 1, target 2): $(|00\rangle + |11\rangle)/\sqrt{2}$. Entanglement made with just two gates!

9. Challenges: Decoherence and Error Correction

- *Qubits lose coherence in microseconds-seconds; circuits need thousands/millions of gates.*
- *Each gate has 0.1-1 % error; errors accumulate (see Unit 3).*
- *QEC encodes logical qubits in many physical qubits; surface code needs ~10^3 physical per logical.*
- *Fault-tolerant computing requires error rates below threshold, fast classical decoding, and fresh ancilla qubits.*
- *Quantum communication faces loss, noise and the lack of practical quantum memory/repeaters.*
- *Interfacing communication & computing: converting stationary qubits (matter) to flying qubits (photons) with high efficiency (transduction) is a research frontier.*

10. Real-World Importance and Future Potential

- *Cybersecurity: migrate to post-quantum cryptography (NIST standards: ML-KEM/Kyber, ML-DSA/Dilithium, SLH-DSA/SPHINCS+); combine with QKD for ultra-sensitive links (banking, defence, government).*
- *Drug & materials discovery: simulate molecules (e.g. FeMoco for fertilizer, battery electrolytes, catalysts).*
- *Optimization: logistics, scheduling, portfolio, energy grids.*
- *Secure cloud: blind quantum computing; quantum-safe cloud services.*
- *Science: tests of fundamental physics, distributed telescope arrays, quantum sensors networks.*
- *Future outlook: hybrid HPC + QPU data centres; 2030s early fault-tolerant machines; global quantum internet by 2040s (optimistic).*

Exam tip / Remember

Timeline hint: 1984 BB84 · 1991 E91 · 1994 Shor · 1996 Grover · 2016 Micius · 2019 'supremacy' · 2022 Nobel Prize (Aspect, Clauser, Zeilinger) · 2024 NIST PQC standards.

10A. Classical Cryptography vs Quantum Threats - Why It Matters

Cryptosystem	Basis	Threat from quantum computer
RSA, Diffie-Hellman, ECC	Factoring / discrete logarithm	Broken by Shor's algorithm
AES-128/256, SHA-2/3	Symmetric & hash	Grover halves effective key length → use AES-256
One-time pad	Random key as long as message	Information-theoretically secure; key distribution is the problem → QKD
Lattice, code, hash-based (PQC)	Hard problems believed quantum-resistant	Standardized by NIST (2024); software-only
QKD	Laws of physics	Secure against any computational power; needs hardware links

'Harvest now, decrypt later'
Adversaries can record encrypted traffic today and decrypt it once a quantum computer exists. Data that must stay secret for 20-30 years (health, defence, diplomatic) is already at risk - so migration to quantum-safe encryption must start now.

» Privacy amplification & one-time pad (QKD post-processing)

- After sifting, Alice and Bob run information reconciliation (e.g. Cascade/LDPC) to fix remaining errors.
- Then privacy amplification: hash the key to a shorter one so that whatever partial information Eve might have is reduced to almost zero.
- Final key used for one-time pad: ciphertext = message XOR key; unbreakable if key is random, used once and secret.

» Quantum random number generators (QRNG)

Classical computers produce pseudo-random numbers by algorithms. A QRNG measures a quantum event - e.g. which output a photon takes at a 50:50 beam splitter, or vacuum fluctuations - giving true randomness certified by physics. Commercial QRNG chips are already in smartphones (e.g. Samsung) and data centres; useful for cryptography, simulations and lotteries.

10B. Quantum Computing Models (Conceptual)

Model	Idea	Examples
Gate-based (circuit)	Sequence of unitary gates on qubits	IBM, Google, IonQ
Measurement-based (one-way)	Prepare big entangled cluster state; measurements drive the computation	Photonic (PsiQuantum, Xanadu)
Adiabatic / Annealing	Slowly evolve a system from simple to problem Hamiltonian; ground state = answer	D-Wave (optimization)
Topological	Information in non-local anyons → inherently error-resistant	Microsoft (research)
Analog quantum simulation	Engineer a quantum system that mimics another	Cold atoms, ion crystals

11. Solved Examples

Example 1 - Sifted key length

Alice sends 1000 photons, bases random. Bob's basis matches ~50 %: sifted key ≈ 500 bits. Sacrificing 100 bits for error check leaves ≈ 400 bits before privacy amplification.

Example 2 - Eavesdropper detection

Intercept-resend by Eve gives QBER ≈ 25 %. If Alice & Bob compare 100 bits, probability that Eve (attacking every photon) is undetected ≈ $(3/4)^{100} \approx 3\times10^{-13}$.

Example 3 - Grover speed-up

Search 1,000,000 items: classical ~ 500,000 checks on average; Grover ≈ $(\pi/4)\sqrt{10^6} \approx 785$ queries.

Example 4 - Fibre loss

200 km at 0.2 dB/km → 40 dB loss → fraction transmitted 10^{-4}. A 1 GHz photon source gives ~10^5 photons/s received.

12. Revision Questions

1. Compare quantum and classical information.
2. Describe the main elements of a quantum communication system.
3. Explain BB84 with an example table. How is eavesdropping detected?
4. Role of entanglement: E91, teleportation, entanglement swapping.

5. *What is a quantum repeater and why is it needed?*
6. *What is the quantum internet? Describe its stages and applications.*
7. *Explain quantum parallelism and its limitation.*
8. *Compare classical and quantum gates; construct a Bell-state circuit.*
9. *Why are decoherence and error correction challenges?*
10. *Discuss the real-world importance of quantum communication and computing.*

13. Short Answer Practice (Model Answers)

Q1. Why is BB84 secure?

Because (i) a single photon cannot be copied (no-cloning), (ii) measuring in the wrong basis randomizes and disturbs the state, and (iii) Alice and Bob can detect the resulting errors by comparing a sample of their key. Any eavesdropping is revealed as an abnormally high QBER.

Q2. QKD vs post-quantum cryptography (PQC)?

QKD: security from physics, needs special hardware & links, limited distance, key exchange only. PQC: software algorithms believed hard even for quantum computers, deployable anywhere, security based on unproven assumptions. Best practice: use both where needed.

Q3. Why can quantum signals not be amplified?

An amplifier would need to copy the photon's unknown state - forbidden by no-cloning. Hence repeaters use entanglement swapping and quantum memories rather than amplification.

Q4. What is quantum advantage vs supremacy?

Supremacy: solving any task a classical machine cannot in reasonable time. Advantage: doing a useful task better than classical methods. Supremacy demos (Google 2019, USTC 2020) are contrived; practical advantage is still pending.

Q5. Why is the Hadamard gate important?

It converts basis states to superpositions ($|0\rangle \rightarrow |+\rangle$) and vice-versa, enabling parallelism and interference. H on all qubits creates the uniform superposition used by Deutsch-Jozsa, Grover and Shor.

Q6. Entanglement swapping in 3 lines

Pairs A-B and C-D are entangled. A Bell measurement on B and C is performed and the result is told to the ends. Now A and D are entangled even though they never interacted - the building block of quantum repeaters.

★ *Unit 4 summary in 5 lines*

1. Quantum communication moves quantum states (photons). 2. QKD (BB84/E91) gives physics-based security. 3. Entanglement → teleportation, swapping, repeaters. 4. Quantum computing = qubits + unitary gates + interference; Shor and Grover are star algorithms. 5. Decoherence and loss are the big hurdles; quantum internet is the long-term goal.